

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC)

April/May 2012

EE220 Electrical Machines & Power Circuits

Date: 05.05.2012, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Draw and explain the internal and external load characteristics of dc series and shunt generator. [07]

- Q - 2 (a) Draw and explain the torque speed characteristics of dc compound motor. [14]
(b) Explain the speed control methods for dc series motor.

OR

- Q - 2 (a) Explain in detail the different types of losses taking place in dc machines. [14]
(b) Draw and explain the working of three point dc shunt motor starter.

- Q - 3 (a) Give classification of different types of single phase induction motor. Explain with necessary diagram construction, working, application of any one. [14]
(b) Draw the equivalent circuit of single phase induction motor using double revolving field and explain the same.

OR

- Q - 3 (a) Explain the maximum demand tariff and two part tariff with examples of each type. [14]
(b) Derive the equation for the most economical power factor.

SECTION - II

Q - 4 Explain the operation of transformer on load. Draw the vector diagram of transformer on load. [07]

- Q - 5 (a) Derive the condition for maximum efficiency of transformer. Define the all-day efficiency of transformer. [14]
(b) State and explain the necessary condition required to be fulfilled for the parallel operation of transformer.

OR

- Q - 5 (a) Draw and explain the torque speed characteristics of three phase induction motor. [14]
(b) Write a technical note on double cage induction motor.

- Q - 6 (a) Differentiate between the types of alternator. [14]
(b) Define the synchronous reactance. Explain the method to determine the synchronous reactance. Define the voltage regulation of alternator.

OR

- Q - 6 (a) Draw the pole mounted substation and explain the different equipment used. [14]
(b) Draw the key diagram of 11kV/400 V indoor substation.